228446: metagranodiorite, Ilbianna Well

(Granitic basement, Capricorn Orogen)

Location and sampling

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MGA Zone 50, 756339E 7502954N

Sampled on 18 October 2017

This sample was collected from the 1885.48 – 1887.40 m depth interval of diamond drillcore DD83FVG1, drilled in 1983–1984 by CRA Exploration Pty Ltd in the Fortescue Valley (Andrew, 1985). The drillhole is located on Marillana Station, about 10.7 km north of Old Woolshed Bore on the Munjina – Roy Hill Road, 6.2 km east-northeast of No. 2 Bore, and 0.7 km north-northeast of Ilbianna Well.

Geological context

The unit sampled is a metagranodiorite, overlain by the Kylena and Jeerinah Formations of the 2775–2629 Ma Fortescue Group and the Marra Mamba Iron Formation and Wittenoom Formation of the 2629–2494 Ma lower Hamersley Group (Thorne and Trendall, 2001). A poorly defined Rb–Sr isochron date of c. 2.4 Ga for altered granodiorite in this drillcore is younger than the expected age of the basement granitic rocks, and possibly represents the time of alteration (Andrew, 1985).

Petrographic description

The sample is an altered, medium-grained biotite metagranodiorite, consisting of about 50–55% plagioclase, 20–25% quartz, 15–20% K-feldspar, 5% biotite, 2% titanite, and accessory myrmekite, allanite, apatite and zircon. Plagioclase is subhedral, equant and tabular, up to 3 mm long, and has been altered by fine sericite, carbonate, chlorite and probably albite. Quartz is anhedral, irregular, and up to 4 mm in size. Some quartz aggregates have serrated intergrown boundaries, and larger quartz grains exhibit strong undulose extinction as well as sub-graining and minor recrystallization. K-feldspar is in part tabular, micropoikilitic and microperthitic, up to at least 6.5 mm long, and smaller (1 mm) anhedral interstitial K-feldspar grains display microcline twinning and small lobes of myrmekite around their margins. Unaltered biotite forms subhedral laths up to 1.5 mm long, although most is significantly altered to chlorite. Euhedral diamond-shape grains composed of oxide and clay minerals after titanite are up to 1 mm long. Brown metamict allanite is subhedral, tabular, and up to 0.2 mm long. The granodiorite has undergone solid-state deformation and recovered at probable low to moderate grade. The rock has subsequently undergone brittle deformation to produce narrow zones of fine brecciation, which have been preferentially altered.

Zircon morphology

Zircons isolated from this sample are colourless to dark brown or opaque and subhedral to euhedral. The crystals are up to 300 µm long, and mainly elongate, with aspect ratios up to 5:1. In cathodoluminescence (CL) images, concentric zoning is ubiquitous, some crystals contain high-uranium, metamict zones, and some appear to contain older cores. A CL image of representative zircons is shown in Figure 2.

Analytical details

This sample was analysed on 5–6 September 2019, using SHRIMP-B. Twelve analyses of the M257 standard obtained during the session indicated an external spot-to-spot (reproducibility) uncertainty of 1.00% (1σ) and a $^{238}U/^{206}Pb^*$ calibration uncertainty of 0.33% (1σ). Calibration uncertainties are included in the errors of $^{238}U/^{206}Pb^*$ ratios and dates listed in Table 1. Common-Pb corrections were applied to all analyses using contemporaneous isotopic compositions determined according to the model of Stacey and Kramers (1975).

Results

Eighteen analyses were obtained from 18 zircons. Results are listed in Table 1, and shown in a concordia diagram (Fig. 3).

Interpretation

The analyses are concordant to moderately discordant (Fig. 3). Four analyses are >7% discordant. The dates obtained from these four analyses (Group D; Table 1) are unreliable, and are considered not to be geologically significant. The remaining 14 analyses can be divided into two groups, based on their $^{207}Pb^*/^{206}Pb^*$ ratios and their locations within the crystals.

Group I comprises seven analyses (Table 1), which yield a weighted mean $^{207}Pb^*/^{206}Pb^*$ date of 2903 ± 4 Ma (MSWD = 0.61).

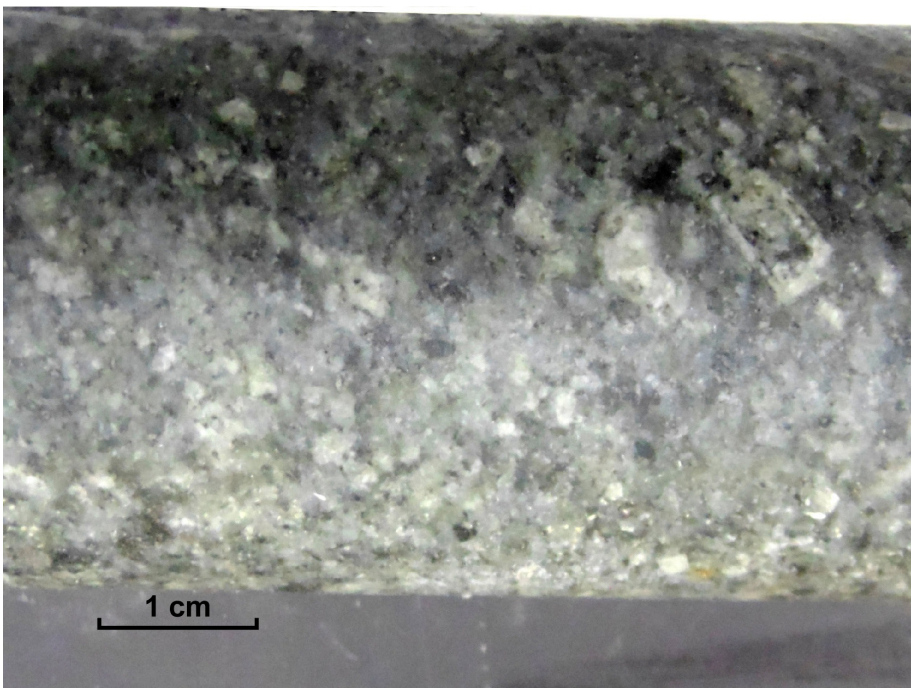

Figure 1. **Drillcore image (wet surface) for sample 228446: metagranodiorite, Ilbianna Well**

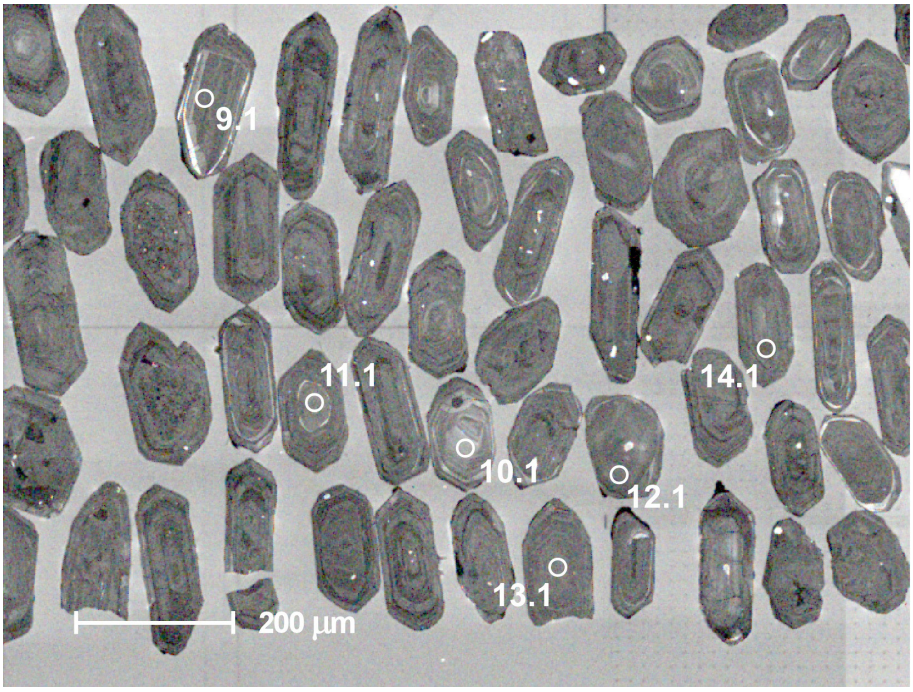

Figure 2. **Cathodoluminescence image of representative zircons from sample 228446: metagranodiorite, Ilbianna Well. Numbered circles indicate the approximate locations of analysis sites**

Table 1. Ion microprobe analytical results for zircons from sample 228446: metagranodiorite, Ilbianna Well

Group ID	Spot no.	Grain. spot	^{238}U (ppm)	^{232}Th (ppm)	$^{232}Th/^{238}U$	f204 (%)	$^{238}U/^{206}Pb$ ± 1σ		$^{207}Pb/^{206}Pb$ ± 1σ		$^{238}U/^{206}Pb^*$ ± 1σ		$^{207}Pb^*/^{206}Pb^*$ ± 1σ		$^{238}U/^{206}Pb^*$ date (Ma) ±1σ		$^{207}Pb^*/^{206}Pb^*$ date (Ma) ±1σ		Disc. (%)
I	6	6.1	323	273	0.85	0.007	1.765	0.021	0.20890	0.00058	1.765	0.021	0.20883	0.00058	2893	28	2897	5	0.1
I	12	12.1	138	160	1.16	0.050	1.791	0.023	0.21010	0.00087	1.792	0.023	0.20965	0.00089	2858	30	2903	7	1.5
I	16	16.1	294	331	1.13	0.073	1.832	0.022	0.21034	0.00092	1.833	0.022	0.20969	0.00093	2806	27	2903	7	3.3
I	8	8.1	387	316	0.82	-0.009	1.868	0.022	0.20960	0.00053	1.868	0.022	0.20968	0.00053	2764	26	2903	4	4.8
I	3	3.1	307	125	0.41	0.132	1.899	0.026	0.21113	0.00067	1.901	0.026	0.20996	0.00070	2725	31	2905	5	6.2
I	17	17.1	226	174	0.77	0.043	1.820	0.027	0.21067	0.00066	1.821	0.027	0.21028	0.00067	2821	34	2908	5	3.0
I	1	1.1	137	127	0.93	0.066	1.803	0.023	0.21094	0.00086	1.804	0.023	0.21035	0.00089	2843	30	2908	7	2.3
X	2	2.1	265	64	0.24	0.026	1.611	0.024	0.23911	0.00069	1.611	0.024	0.23888	0.00070	3113	38	3113	5	0.0
X	9	9.1	162	84	0.52	0.271	1.614	0.020	0.25068	0.00085	1.618	0.020	0.24831	0.00095	3102	31	3174	6	2.3
X	5	5.1	214	120	0.56	0.234	1.587	0.019	0.25118	0.00077	1.590	0.019	0.24913	0.00084	3145	31	3179	5	1.1
X	11	11.1	133	105	0.79	0.130	1.599	0.021	0.25062	0.00157	1.601	0.021	0.24948	0.00159	3128	33	3182	10	1.7
X	18	18.1	268	198	0.74	0.041	1.593	0.022	0.25150	0.00105	1.594	0.022	0.25114	0.00105	3139	35	3192	7	1.7
X	10	10.1	96	68	0.71	0.056	1.525	0.021	0.26366	0.00112	1.525	0.021	0.26317	0.00113	3250	36	3266	7	0.5
X	7	7.1	167	63	0.38	0.027	1.507	0.025	0.27143	0.00091	1.507	0.025	0.27119	0.00091	3281	44	3313	5	1.0
D	13	13.1	443	429	0.97	0.019	2.231	0.025	0.19602	0.00051	2.232	0.025	0.19585	0.00051	2387	23	2792	4	14.5
D	14	14.1	431	351	0.81	0.022	1.969	0.025	0.20386	0.00047	1.970	0.025	0.20366	0.00047	2647	28	2856	4	7.3
D	15	15.1	299	203	0.68	0.061	1.968	0.023	0.20433	0.00058	1.969	0.023	0.20378	0.00059	2648	26	2857	5	7.3
D	4	4.1	131	85	0.65	0.111	1.784	0.023	0.24917	0.00101	1.786	0.023	0.24819	0.00105	2866	31	3173	7	9.7

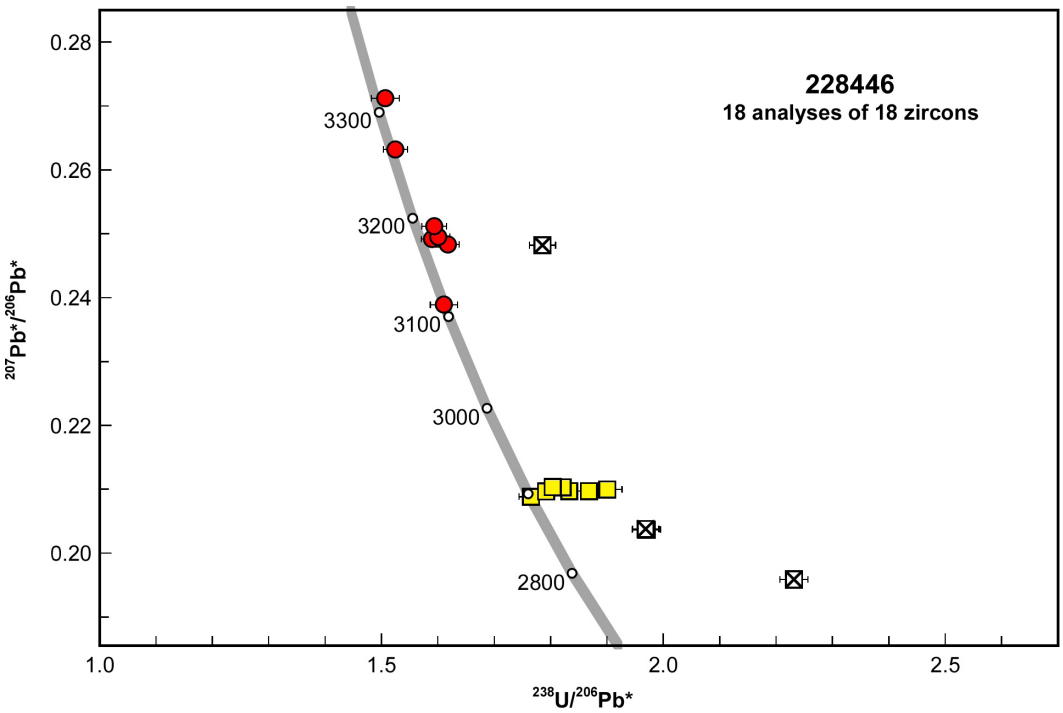

Figure 3. U–Pb analytical data for zircons from sample 228446: metagranodiorite, Ilbianna Well. Yellow squares indicate Group I (magmatic zircons); red circles indicate Group X (xenocrystic zircon cores); crossed squares indicate Group D (discordance >7%)

Group X comprises seven analyses of zircon cores (Table 1), which yield $^{207}Pb^*/^{206}Pb^*$ dates of 3313–3113 Ma. Four analyses in Group X yield a weighted mean $^{207}Pb^*/^{206}Pb^*$ date of 3181 ± 12 Ma (MSWD = 1.4).

The date of 2903 ± 4 Ma for the seven analyses in Group I is interpreted as the igneous crystallization age of the granodiorite. The dates of 3313–3113 Ma for the seven analyses in Group X are interpreted as the ages of inherited zircons. The date of 3181 ± 12 Ma for four analyses in Group X is interpreted as the age of an inherited component.

References

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Stacey, JS and Kramers, JD 1975, Approximation of terrestrial lead isotope evolution by a two-stage model: Earth and Planetary Science Letters, v. 26, p. 207–221.

Thorne, AM and Trendall, AF 2001, Geology of the Fortescue Group, Pilbara Craton, Western Australia: Geological Survey of Western Australia, Bulletin 144, 249p

Recommended reference for this publication

Wingate, MTD, Lu, Y, Fielding, IOH and Howard, HM 2021, 228446: metagranodiorite, Ilbianna Well; Geochronology Record 1731: Geological Survey of Western Australia, 4p.

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